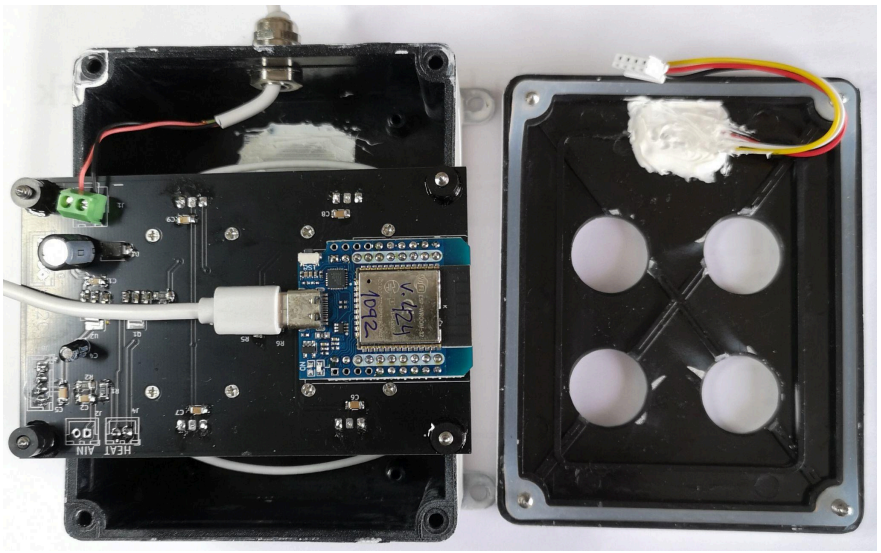
TESS-W4C Firmware Upgrade

To update the firmware you need a micro or C type USB cable, depending on version., the .bin file and the upload program. It is necessary to open the photometer to connect the USB cable.

1- Remove the cover by unscrewing the top four screws.

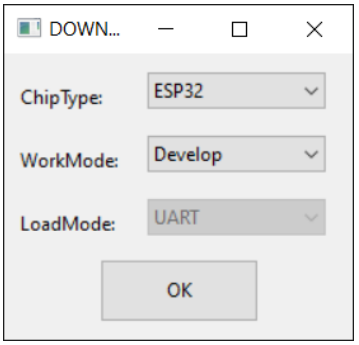


2- Unscrew the PCB to access the components, and connect the USB cable to the module EP32.

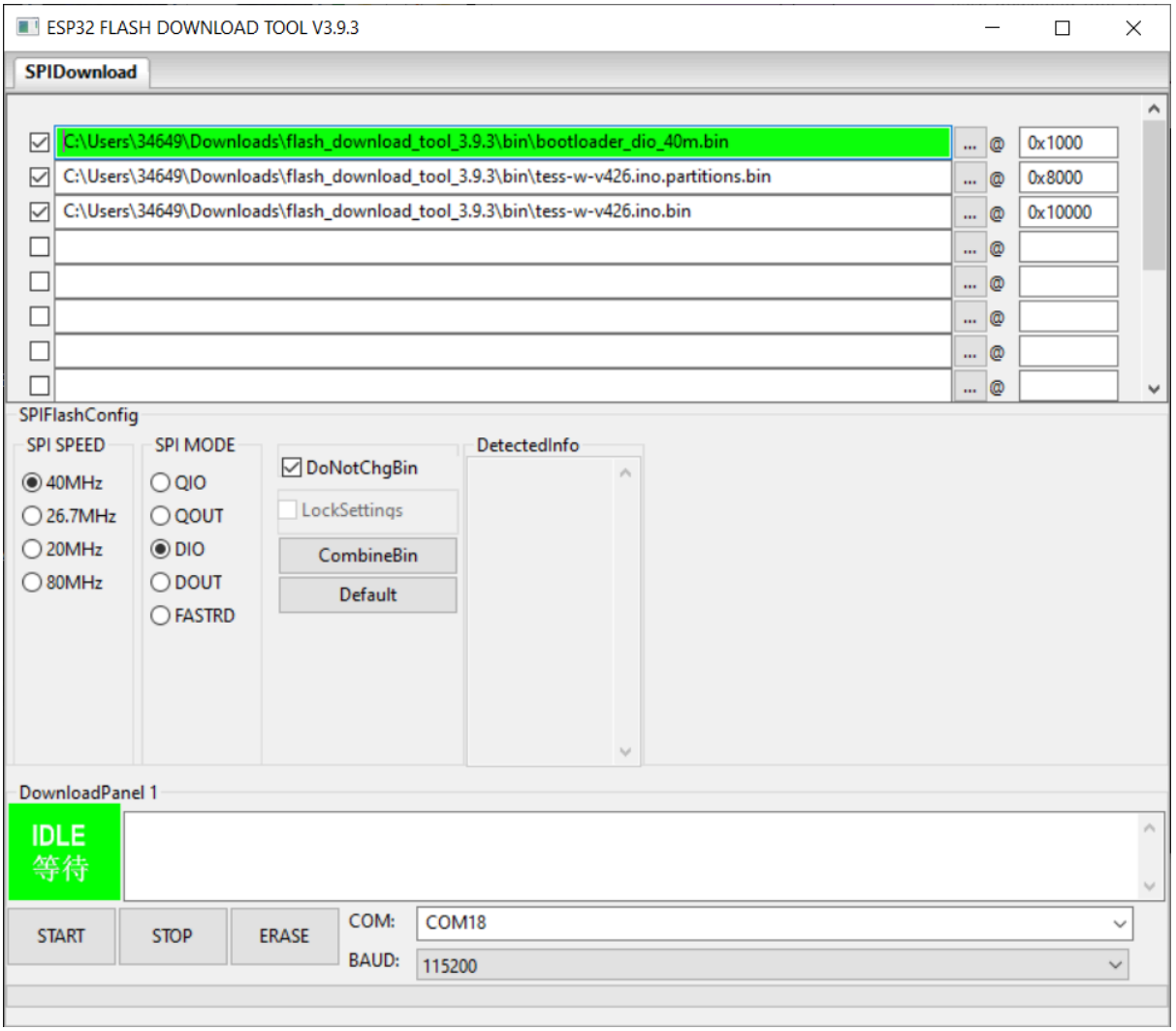


3- Download and unzip the update package.
https://observatorioremoto.com/TessW4C_downloadtool.zip
It contains the update tool, the binary files to be loaded and a small terminal program called Termite for testing.
Before starting the update, use Termite and check how the photometer works. This test is described in the final section.
.

4- Launch the upload program by clicking on the executable *flash_download_tool_3.9.3.exe*
Select Chip Type: ESP32 and press OK.

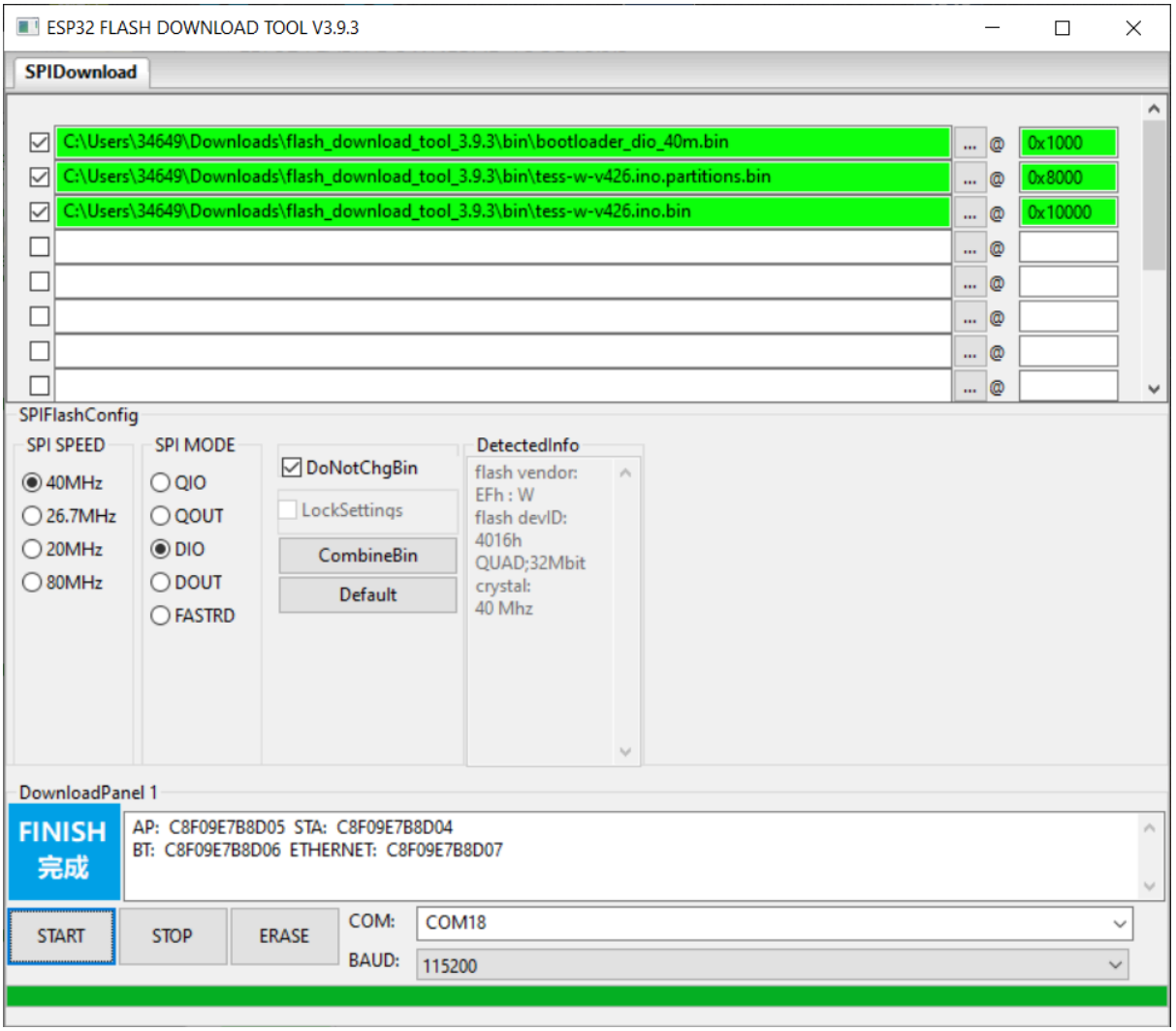


The main window appears showing three binary files located in the \bin folder.
The path to the files can be chosen from the button on the right of the file. They should be changed if they are on another path.



Selects the COM port assigned by Windows to the sensor and press START button.

5- Final status if all went well.

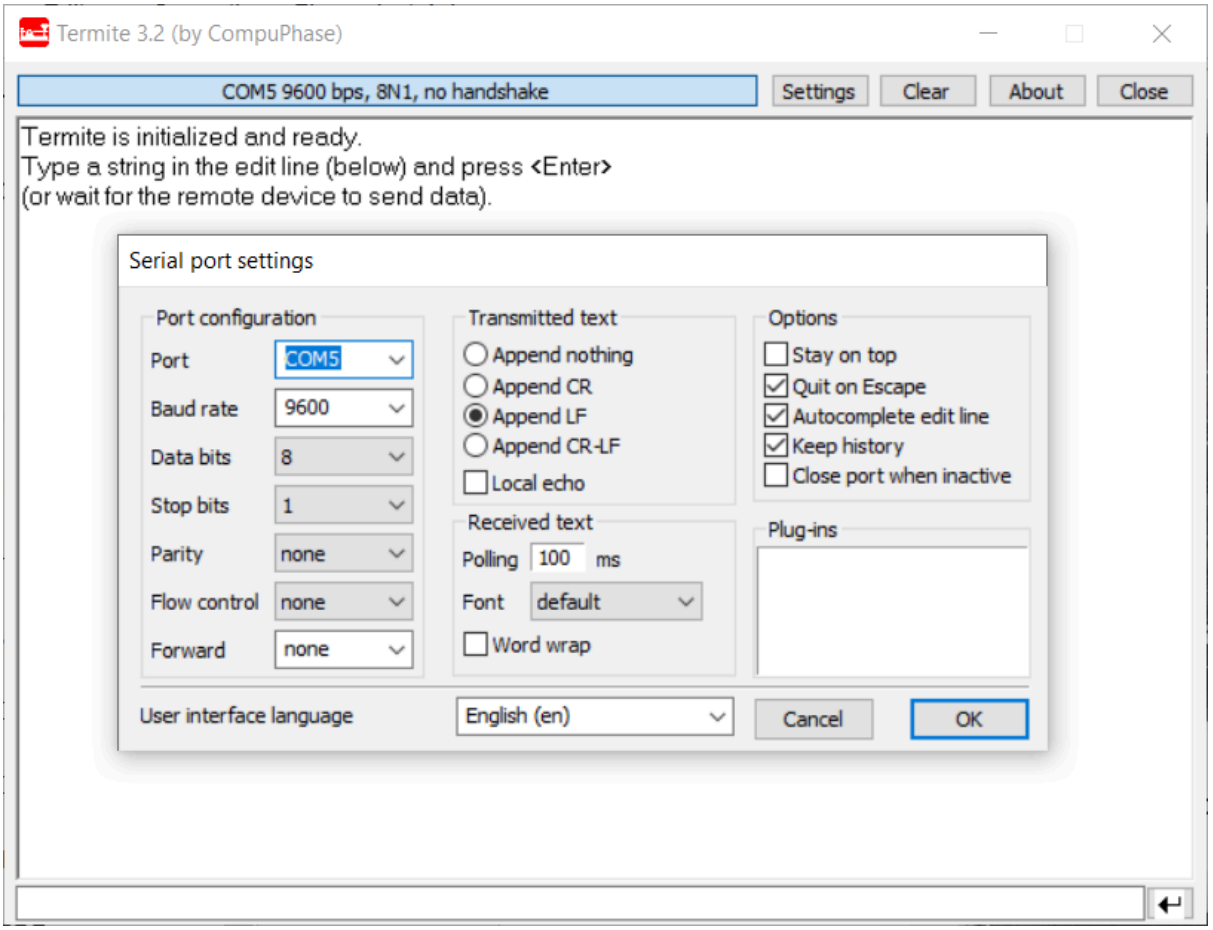


Close the Download Tool.

Testing TESS-W4C internally.

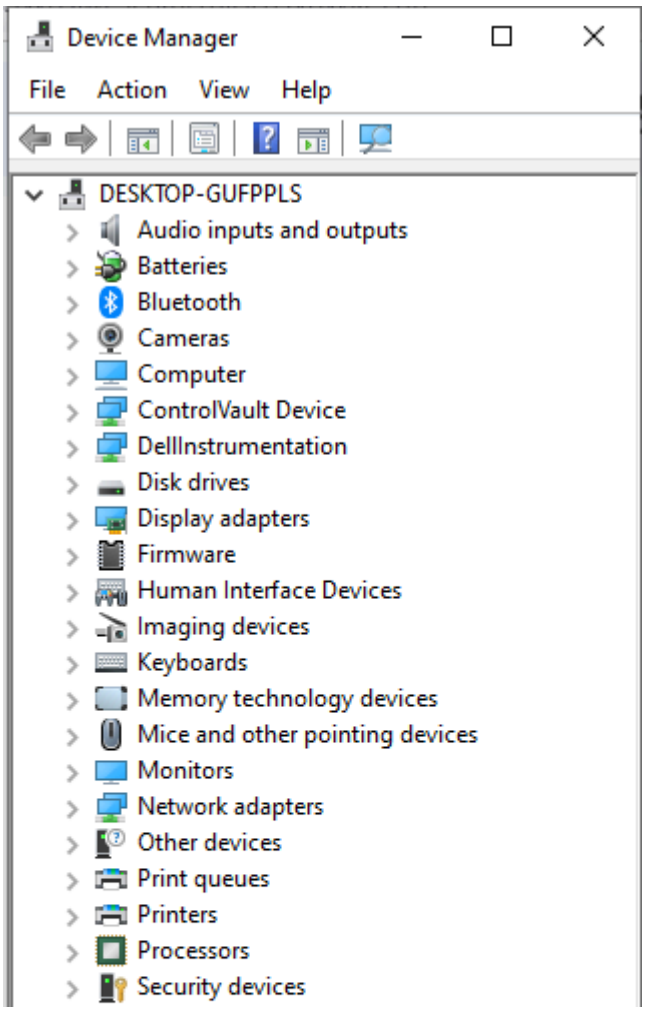
Using a USB cable to micro or C, we can connect internally to the photometer and with the program Termite.exe observe its operation.

After clicking Termite.exe, go to Settings and select the COM port assigned to your TESS.

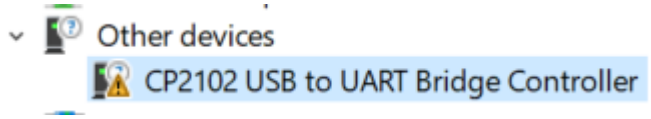


It may happen that the first time we connect the photometer, Windows does not find on the internet the driver for the USB to serial adapter of the photometer. Then we have to search for it and download it manually.

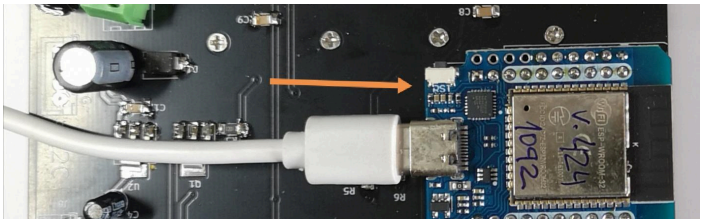
To download the driver we first need to know which chip is in our photometer.
With the photometer connected to our PC, open the **Device Manager:**



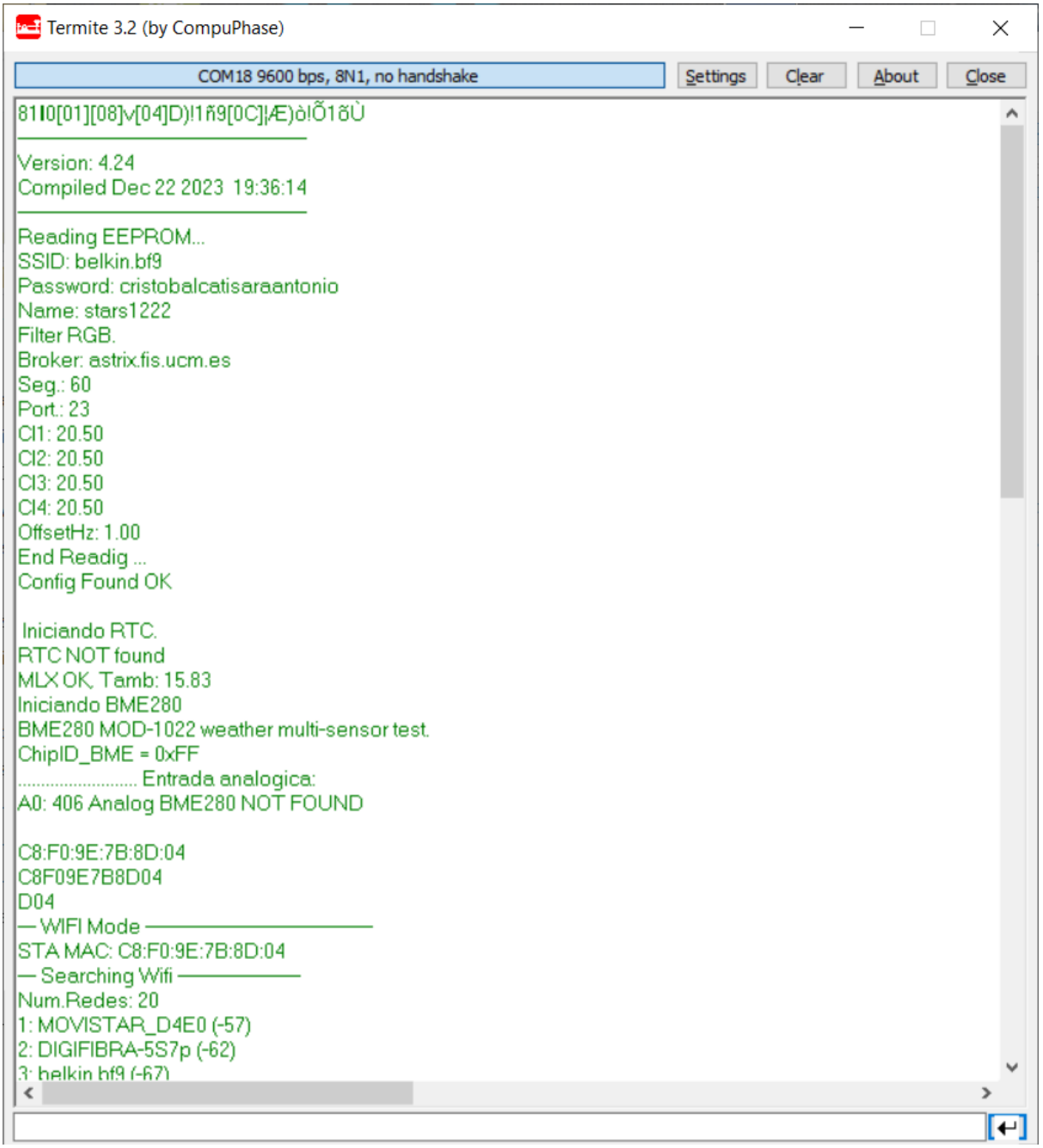
and cllick on **Other devices**, it will show the chip installed in Tess. It is for that chip that we need to look for the driver on internet:

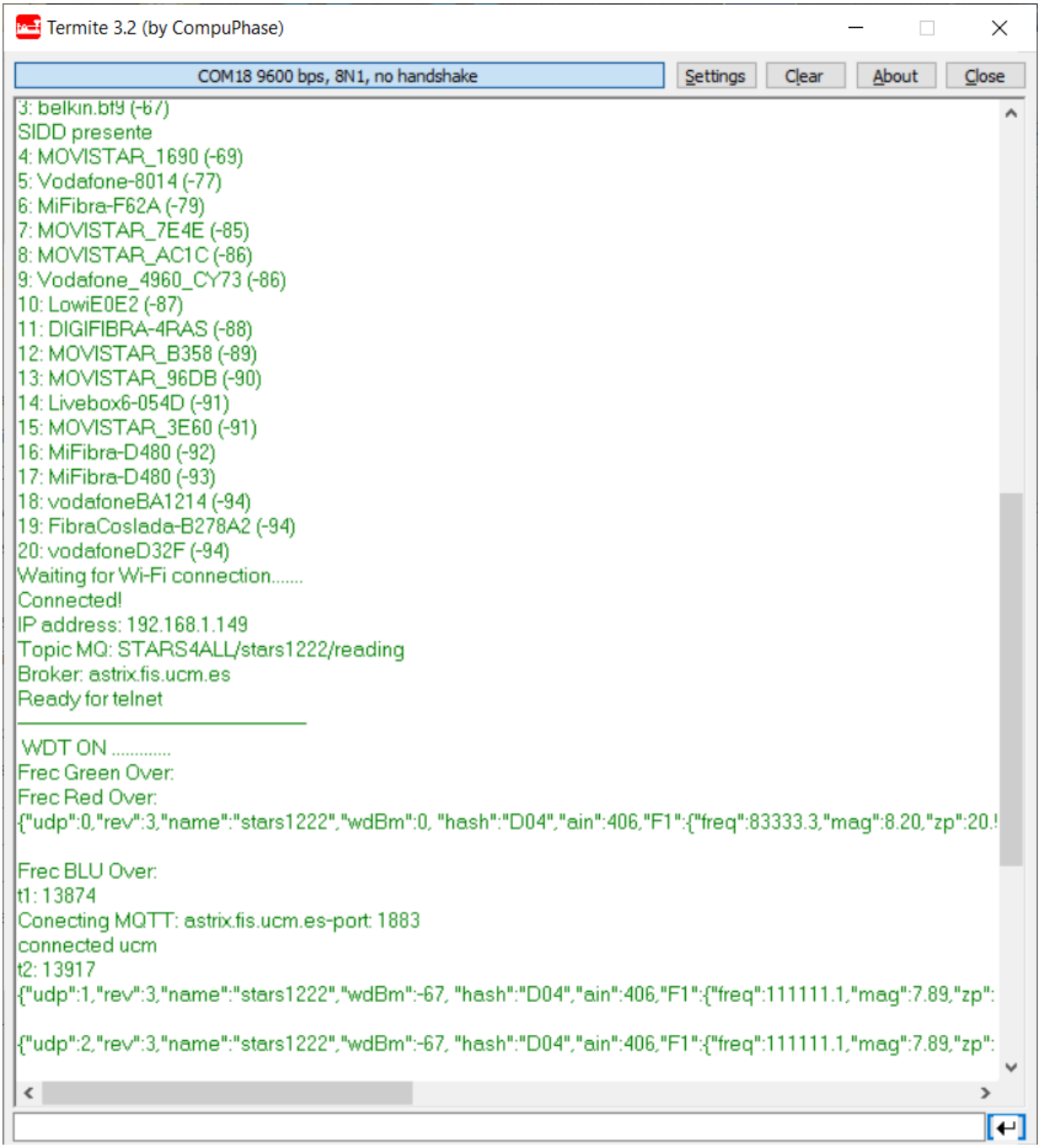


Once the USB driver is installed, open Termite, and reset the module by pressing the small button on its board:



you will see the Tess sensor configuration, the start of the program and finally the readings. In a module with configuration prior to the update, we can see something like this:


Termite 3.2 (by CompuPhase)
COM18 9600 bps, 8N1, no handshake
Settings
Clear
About
Close
81I0[01][08]v[04]D)I1ñ9[0C]¦Æ)ò!Õ1õÙ
Version: 4.24
Compiled Dec 22 2023 19:36:14
Reading EEPROM...
SSID: belkin.bf9
Password: cristobalcatisaraantonio
Name: stars1222
Filter RGB.
Broker: astrix.fis.ucm.es
Seg.: 60
Port.: 23
CI1: 20.50
CI2: 20.50
CI3: 20.50
CI4: 20.50
OffsetHz: 1.00
End Readig ...
Config Found OK
Iniciando RTC.
RTC NOT found
MLX OK, Tamb: 15.83
Iniciando BME280
BME280 MOD-1022 weather multi-sensor test.
ChipID_BME = 0xFF
..... Entrada analogica:
A0: 406 Analog BME280 NOT FOUND
C8:F0:9E:7B:8D:04
C8F09E7B8D04
D04
— WIFI Mode ————————————
STA MAC: C8:F0:9E:7B:8D:04
— Searching Wifi ————————
Num.Redes: 20
1: MOVISTAR_D4E0 (-57)
2: DIGIFIBRA-5S7p (-62)
3: belkin.bf9 (-67)



In order to read the information, we can stop and start the data by clicking on the top bar, which indicates the COM port.

If the TESS had no previous configuration or had been deleted with a Full Reset we will see this:

